

Santiago Vázquez Villarreal

Full-Stack Engineer · Spanish / English / French

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Profile

Full-stack engineer with a year and a half shipping production systems for municipal clients. I'm lead engineer at CityFront AI. I designed our embeddable AI chat widget and led our backend rewrite from AWS serverless to NestJS with Postgres in Docker, which cut new-client onboarding from about two weeks to under three days. I care a lot about systems the next engineer can onboard onto without help.

Experience

CityFront AI

Full-Stack Engineer

Remote (Dallas, TX)

Jan 2025 – Present

- Took our tenant deployment from per-client custom work to a config-driven flow. Time-to-live for a new municipality dropped from **about two weeks to under three days**.
- Pitched and led our backend migration off AWS serverless onto NestJS with Postgres in Docker. Right-sized for a small startup, fully local dev loop. New engineers boot the whole stack in **under five minutes**.
- Designed and shipped **AskEcho**, our embeddable AI chat widget. A **~5KB** vanilla-TypeScript loader, a sandboxed iframe with Shadow-DOM and cross-origin postMessage, a SvelteKit UI with SSE token streaming. Drops into any CMS via a single `<script>` tag.
- Built the multi-tenant RAG platform behind it. Django, Postgres with pgvector, Pinecone, OpenAI. Per-tenant vector namespaces, intent-based workflow routing, Cognito JWKS auth. Handles **low-thousands of resident interactions a month** right now.
- Built our integration layer to legacy work-order systems like CityWorks. Adapter pattern, so adding the next vendor is **a few hundred lines**. **AES-256-GCM** for credentials, **HMAC-signed** webhooks with dedupe and audit. Cities trust us with their data, so the unglamorous parts matter.
- Own **4 production apps** across pnpm monorepos: the widget, the RAG platform, our McKinney resident portal on Next.js 16, and the staff/admin dashboard. Three-environment CI/CD on GitHub Actions, Playwright, Vitest, and Jest at **98%+ coverage**, Sentry on both ends.
- Mentor a junior developer and an intern on our 3-person team. The junior is moving from frontend-only into security and AWS-architecture work on AI products. The intern started on a Svelte landing page and is now shipping a React Native mobile app on FastAPI.

John Deere

Embedded Software Developer Intern

Monterrey, Mexico

Feb 2024 – Aug 2024

- Wrote onboard diagnostic alerts in C++ for embedded tractor systems to catch early signs of mechanical failure. The bar for correctness was high; you can't push a hotfix to a tractor.
- Contributed to **3 release bundles** across production sprayer and planting platforms. Optimized memory-constrained C++ modules and validated everything through hardware-in-the-loop testing.

Projects

- **Wireless Pick-to-Light System**. Coppel Capstone (Project Lead, Mar–Jun 2025). ESP32 modules over MQTT, with a FastAPI backend on PostgreSQL coordinating real-time hardware-to-workflow communication. Projected hardware and install cost **beat the legacy wired vendor solution** while collapsing the operator interface to a single LED and button.
- **VVF**. Single-tenant ERP for a Monterrey garment manufacturer (real client). SvelteKit, TypeScript, Supabase. Deployed to Cloudflare Pages. Phased delivery across facilities, materials, products, BOMs, and per-facility inventory.
- **Piggy**. Personal-finance app. Monorepo with a React Native (Expo SDK 54) mobile app and a FastAPI backend on PostgreSQL. Clerk JWT auth, NativeWind, TanStack Query, Zustand, PgBouncer in front of Postgres.
- **Knot**. Wedding-planning SaaS for couples and their guests. SvelteKit frontend with a FastAPI backend on PostgreSQL. Automatic RSVP, scheduled-payment calendar, guest and people management, registry-list distribution.

Education

Tecnológico de Monterrey

B.S. Innovation & Development Engineering, *Summa Cum Laude*, GPA 4.0

Monterrey, Mexico

Aug 2021 – Jul 2025

IESEG School of Management

Exchange Semester, Business & Innovation, GPA 4.0

Lille, France

Aug 2024 – Dec 2024

Technical Skills

- **Programming**: TypeScript, JavaScript, Python, C++, Go, SQL
- **Web**: React, Next.js (App Router), SvelteKit, Svelte 5, Astro, TailwindCSS, Vite; NestJS, Django, FastAPI, Node.js; REST, GraphQL (AppSync), SSE, OpenAPI
- **Data & AI**: PostgreSQL, Prisma, DynamoDB, pgvector, Pinecone, OpenAI / LLM tool-use, RAG architectures
- **Cloud & Infra**: AWS (CDK, AppSync, Cognito, Lambda, DynamoDB, S3, CloudFront, App Runner, Batch, RDS), Docker / Docker Compose, Cloudflare, Supabase
- **Quality & Ops**: Playwright, Vitest, Jest, GitHub Actions CI/CD, Sentry
- **Interests**: Rock climbing, golf, music performance, long-form writing